

Fossil fuels

Learning outcomes

By the end of this section you should be able to:

- Evaluate the amount of carbon dioxide added to the atmosphere when different fuels burn.
- Compare the relative amounts of energy released per unit mass of different fossil fuels and their tendency to undergo incomplete combustion.
- Identify the relationship that exists between carbon dioxide levels and the greenhouse effect.

Not all fuels are pure substances like the example of the combustion of propane given in the previous section. Some fuels are mixtures of many compounds. Fossil fuels can contain a variety of compounds that formed through the decomposition of prehistoric organisms over time, deep within the Earth's crust. The use of fossil fuels for heat and energy is widespread in homes and businesses. How might the combustion of fossil fuels comprising a range of compounds differ compared to the burning of those that contain only one fuel source, such as propane?



Figure 1. Fossil fuels provide nearly 85% of the world's total energy.

Credit: Schropfschop, Getty Images

Fossil fuels

Fossil fuels are subdivided into a variety of categories based on the type of prehistoric organism they formed from and their physical and chemical properties. Fossil fuels are largely divided into three main categories; coal, crude oil, and natural gas (**Table 1**). The components of fossil fuel mixtures can be separated by their boiling point in a process called fractional distillation.

Table 1. Features of fossil fuels.

Type of fossil fuel	State at room temperature	Composition
Coal	Solid	Mainly carbon, traces of hydrogen, oxygen, nitrogen, sulfur and metals.
Crude oil	Liquid	Mixture of medium to long chain hydrocarbons, traces of sulfur, nitrogen and oxygen and metals.
Natural gas	Gas	Mostly methane, traces of other small hydrocarbons, water and carbon dioxide.

Advantages and disadvantages of fossil fuels

Fossil fuels are highly exothermic when combusted and are relatively easy to transport, making them convenient to use in many industries. Also, fossil fuels are abundant in many countries and are relatively inexpensive to obtain compared to the preliminary set-up cost of other types of energy resources, such as nuclear power. Fossil fuels contribute to the overall welfare of people and the economy by providing jobs, income and taxes for many countries and regions. They also support other industries, such as agriculture, manufacturing and power generation.

Conversely, fossil fuels have many disadvantages. The mining, transportation, and processing of fossil fuels are damaging to both local and global environments, further enhanced by accidents such as oil spills, pipeline leaks and coal mine explosions. The burning of fossil fuels produces substances that are harmful to human health and the environment, contributing to global warming, respiratory problems and the acidification of lakes and soil. Since fossil fuels cannot be synthesised, our planet has exhaustible stores of these non-renewable resources, making extraction and processing more challenging.

Making connections

Due to the combination of elements and compounds found in many fuels, the combustion of fuels like coal, biofuel, gasoline, and wood can produce a variety of carbon oxides, sulfur oxides (SO_x) and oxides of nitrogen (NO_x). The contribution of these products to the acidity of rain and other forms of precipitation is further described in [section S3.1.5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-and-nonmetallic-continuum-id-44695/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-and-nonmetallic-continuum-id-44695/>).

Greenhouse effect

Of major importance is the environmental impact of the combustion of fossil fuels. The products of combustion – carbon oxides and other oxides – are released into atmosphere. With the widespread use of fossil fuels, what effect does this have?

Examine **Figure 2** below. It shows the concentration of CO₂ in the atmosphere over a prolonged period of time. What do you notice about the data?

There is a regular cyclic pattern in the data dating back over thousands of years. The CO₂ levels have naturally risen and fallen throughout history. Look at the far right of the chart where the data shows a rapid rise in CO₂ levels. This rapid rise has been determined to be mostly caused by human's use of fossil fuels, producing the highest levels of CO₂ in the atmosphere.

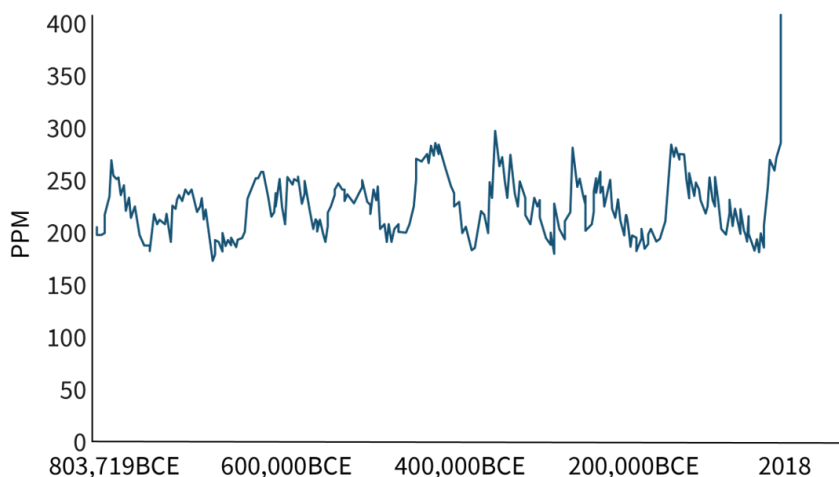


Figure 2. Atmospheric carbon dioxide levels through history.

 More information for figure 2

The image is a line graph depicting atmospheric carbon dioxide levels over time, extending back to 803,719 BCE and continuing to the year 2018. The X-axis represents time in years, marked at intervals of 803,719 BCE, 600,000 BCE, 400,000 BCE, 200,000 BCE, and 2018. The Y-axis represents CO₂ levels in parts per million (PPM), ranging from 0 to 400. Throughout history, the graph shows a regular cyclic pattern, with levels consistently rising and falling. A notable feature is the sharp rise in CO₂ levels at the far right of the chart, indicating a significant increase in recent times, attributed mainly to human activity and fossil fuel consumption.

[Generated by AI]

Now examine **Figure 3**, which contains the average global temperature over the last two centuries. What similarities do you see with **Figure 2** above?

There is a cyclic pattern to the global temperature, just as with the CO₂ levels. There is also a rapid increase in global temperature in more recent years. It appears that the two variables – CO₂ levels and global temperature – are related. Why do you think this is so?

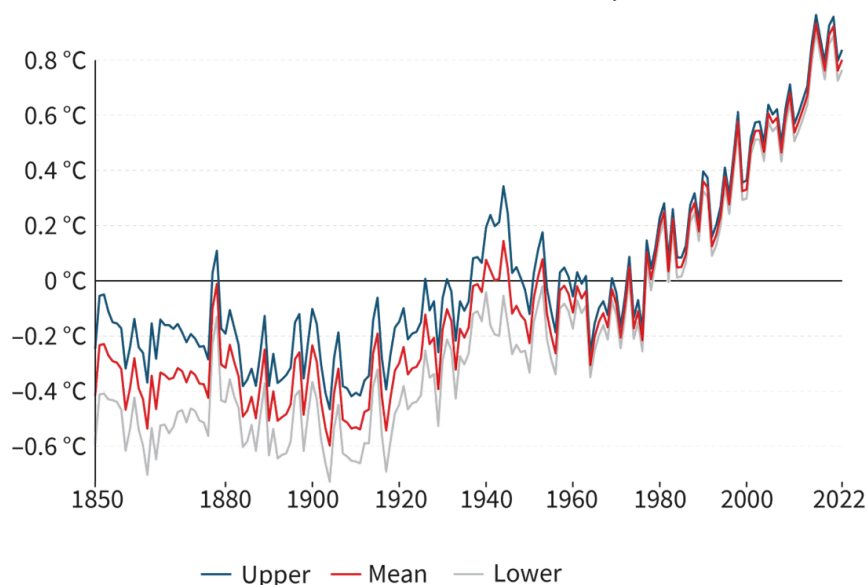


Figure 3. Global temperature through history.

[More information for figure 3](#)

The graph depicts global temperature changes from 1850 to 2022. The Y-axis indicates temperature deviations in degrees Celsius ranging from -0.6°C to $+0.8^{\circ}\text{C}$. The X-axis spans the years 1850 to 2022. Three lines are shown: Upper, Mean, and Lower. From 1850 to around 1930, temperatures showed fluctuations with several peaks and troughs but remained below 0°C . From 1930 onwards, temperatures began rising, with notable peaks around 1940, 1980, and reaching a rapid increase after 2000. By 2022, temperatures reached around $+0.8^{\circ}\text{C}$, indicating a clear upward trend in global temperatures over the years.

[Generated by AI]

The greenhouse effect is a phenomenon that occurs when certain gases, such as carbon dioxide, methane or nitrogen oxides, absorb infrared (IR) radiation that has been emitted from the Earth's surface after it has been heated by the Sun's rays. The greenhouse gases then emit the IR radiation in all directions so the net effect is that a slightly higher proportion of the solar energy is trapped within the atmosphere, leading to higher average global temperatures. Since the combustion of fossil fuels releases carbon dioxide gas, their continued use contributes to the greenhouse effect.

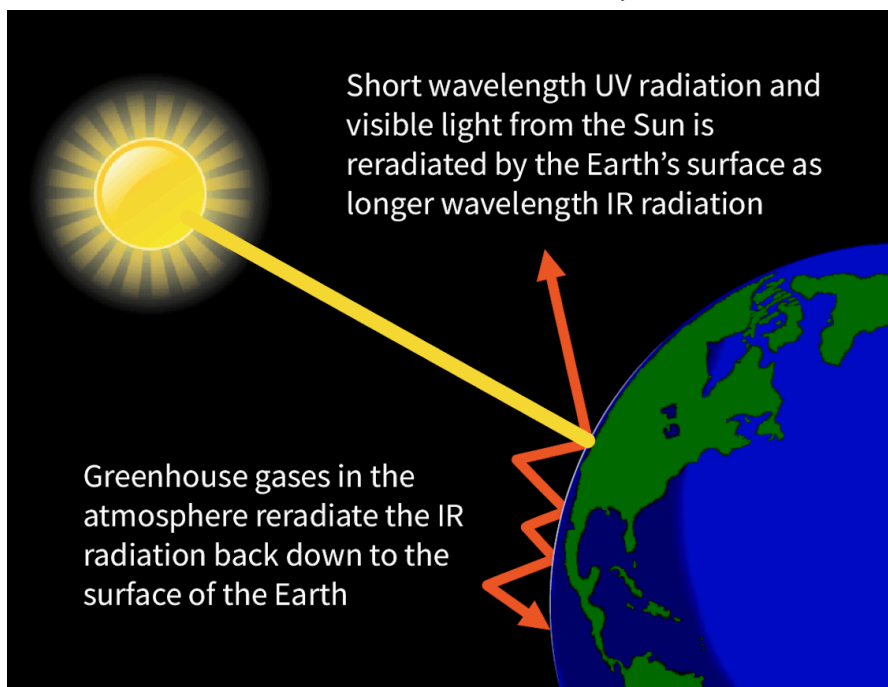


Figure 4. The relationship between atmospheric carbon dioxide levels and the greenhouse effect.

Source: adapted from 'Climate Change Schematic' [↗](#)

(https://commons.wikimedia.org/wiki/File:Climate_Change_Schematic.svg)

by Efbrazil, licensed under CC BY-SA 4.0 [↗](#)

(<https://creativecommons.org/licenses/by-sa/4.0/deed.en>)

[🔍](#) More information for figure 4

The image consists of a diagram illustrating the greenhouse effect. It shows a representation of the Earth with solar radiation depicted as a thick yellow line coming from the sun towards the Earth. Some of this radiation is re-radiated back into space as illustrated by a smaller red arrow noting 'Short wavelength UV radiation and visible light from the Sun is re-radiated by the Earth's surface as longer wavelength IR radiation'. Some radiation is also re-emitted towards Earth, as indicated by a zig-zagging red arrow noting 'Greenhouse gases in the atmosphere re-radiate the IR radiation back down to the surface of the Earth'.

International Mindedness

Fossil fuels like coal and crude oil naturally contain sulfur and nitrogen atoms due to the residues present in the proteins of the pre-historic biological materials that formed these deposits. When fossil fuels are combusted, they produce not only carbon oxides, and water vapour, but also sulfur oxides (SO_x) and oxides of nitrogen (NO_x). Both nitrogen dioxide and sulfur dioxide contribute to acid deposition as is seen in [section S3.1.5](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-and-nonmetallic-continuum-id-44695/>). Due to prevailing winds, the air pollution produced in one country or region flows into other countries and territories. The products of the combustion of fossil fuels is a global issue and one we must address as a global community.

Amount of carbon dioxide produced

All fossil fuels release carbon dioxide when they undergo combustion, but the amount varies depending on the fuel type.

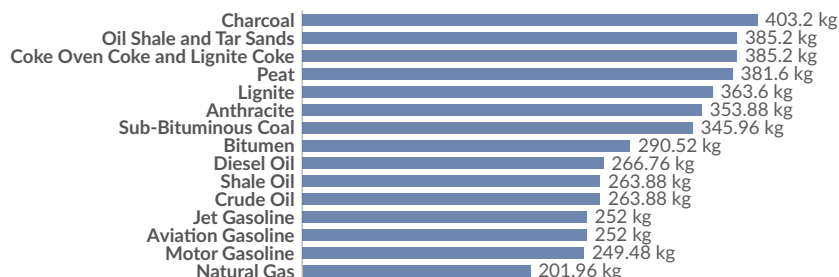
Carbon dioxide emissions factors

Emissions factors quantify the average CO₂ output per unit of energy. They are measured in kilograms of CO₂ per megawatt-hour (MWh) of energy from various fossil fuel sources.

Our World
in Data

Table Chart

Edit



Data source: IPCC - Emission Factor Database

(2025) - [Learn more about this data](#)

OurWorldinData.org/fossil-fuels | CC BY



[Explore the data](#) →

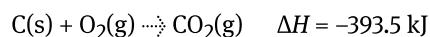
Figure 5. The amount of carbon dioxide released per unit energy of fuel.

More information for figure 5

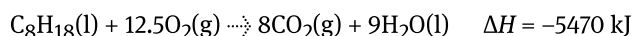
The interactive helps students assess the environmental impact of burning various fuels by assessing CO₂ emissions, comparing energy outputs and combustion efficiencies, and examining how these factors contribute to the greenhouse effect.

It features a horizontal bar chart that displays the carbon dioxide emission factors, measured in kilograms per megawatt-hour, for various fossil fuel sources. The y-axis represents values in kilograms. The x-axis represents various fossil fuel sources. The values, in decreasing order, are provided in a separate tab titled table.

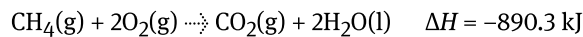
Coal, consisting mainly of carbon, produces a high quantity of carbon dioxide during its combustion, as shown in the reaction equation. Since there is only one product formed in this reaction, the quantity of CO₂ is high compared to the amount of energy released.



Crude oil, a mixture of longer chain hydrocarbons, produces carbon dioxide and water during combustion, as shown in the reaction equation below for octane, one of the components of crude oil. Since there are two products formed, the amount of CO₂ produced per unit of energy released is lower than coal.



Natural gas, consisting mostly of methane, also produces carbon dioxide and water, as shown in the reaction equation below. Just as with crude oil above, the formation of two products lowers the ratio of CO₂ formed per unit of energy. Natural gas has a lower carbon to hydrogen ratio compared to longer hydrocarbon chains; 1 C: 4 H in methane, compared to 1 C: 2.25 H in octane, resulting in less CO₂ released per unit energy compared to crude oil.



These three fuel sources can be compared with the amount, in mol (or by volume), of CO_2 produced per unit of heat (**Table 2**).

Table 2. CO_2 per unit of energy from fuel sources.

Type of fuel	Reaction equation	Mol CO_2 per MJ of energy
Coal	$\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$	$\frac{1 \text{ mol CO}_2}{-393.5 \text{ kJ}} \times 1000 = 2.54 \text{ mol CO}_2 \text{ MJ}^{-1}$
Crude oil	$\text{C}_8\text{H}_{18}(\text{l}) + 12.5\text{O}_2(\text{g}) \rightarrow 8\text{CO}_2(\text{g}) + 9\text{H}_2\text{O}(\text{l})$	$\frac{8 \text{ mol CO}_2}{-5074 \text{ kJ}} \times 1000 = 1.58 \text{ mol CO}_2 \text{ MJ}^{-1}$
Natural gas	$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$	$\frac{1 \text{ mol CO}_2}{-890.3 \text{ kJ}} \times 1000 = 1.12 \text{ mol CO}_2 \text{ MJ}^{-1}$

Therefore, coal produces the greatest amount of CO_2 per unit energy, followed by crude oil and then natural gas.

If the reaction occurs at standard temperature and pressure, we can use the mole ratios and molar volume of gas produced in the reaction.

Table 3. Molar volume of CO_2 produced from fuel sources.

Type of fuel	Amount CO_2 produced	Mass of fuel	Volume per fuel
Coal	$1 \text{ mol} \times 22.7 \text{ dm}^3 \text{ mol}^{-1} = 22.7 \text{ dm}^3$	$1 \text{ mol of C} = 12.01 \text{ g}$	$\frac{22.7}{12.01} = 1.8$
Crude oil	$8 \text{ mol} \times 22.7 \text{ dm}^3 \text{ mol}^{-1} = 181.6 \text{ dm}^3$	$1 \text{ mol C}_8\text{H}_{18} = 114.26 \text{ g mol}^{-1}$	$\frac{181.6}{114.26} = 1.6$

Type of fuel	Amount CO ₂ produced	Mass of fuel	Volume per fuel
Natural gas	$1 \text{ mol} \times 22.7 \text{ dm}^3 \text{ mol}^{-1} = 22.7 \text{ dm}^3$	$1 \text{ mol CH}_4 = 16.05 \text{ g}$	$\frac{22.7}{16.05} = 1.4$

Therefore, coal also produces the largest volume of carbon dioxide per unit mass of fuel, followed by crude oil and natural gas produces the least.

Theory of Knowledge

You can see real time air quality values for many cities in the world at <https://waqi.info/>  (<https://waqi.info/>). The laws promoting and enforcing clean air vary from country to country, but the air is a shared resource. Who has the responsibility for the quality of the air we breathe?

Energy released per unit mass

All fossil fuels release energy when they undergo combustion, but the amount varies depending on the fuel type. The amount of energy can be compared per mole in thermochemical equations for a specific substance undergoing combustion. Since the fossil fuel types are mixtures and vary in composition, the amount of energy released per unit mass of fuel is often used for comparison, and these values can vary as well. As you can see in **Table 3**, coal produces the least amount of energy per unit mass, followed by crude oil and natural gas.

Table 3. Energy per unit mass from fuel sources.

Type of fuel	Energy per unit mass (MJ kg ⁻¹)
Coal	29.3
Crude oil	42.3

Type of fuel	Energy per unit mass (MJ kg ⁻¹)
Natural gas	50.0

Tendency for incomplete combustion

As you have seen, in [section R1.3.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/incomplete-combustion-of-organic-compounds-id-45275/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/incomplete-combustion-of-organic-compounds-id-45275/>) incomplete combustion reactions lead to the formation of carbon monoxide gas and not carbon dioxide gas. All forms of fossil fuels are at risk for incomplete combustion if the air to fuel ratio is not optimised. Consequences of incomplete combustion are:

- less energy released per mole of fuel in comparison to complete combustion reactions
- harmful by-products, including carbon monoxide gas and carbon (soot) particulates
- unreacted fuel can pose a fire/explosion hazard.

Coal has the greatest tendency to undergo incomplete combustion due to the presence of impurities such as nitrogen, sulfur and other metals that also react with oxygen. Since these impurities consume oxygen, there may not be enough oxygen left to react with the carbon in the coal, leading to incomplete combustion.

Crude oil and natural gas mostly consist of hydrocarbons – crude oil with longer chains and natural gas mostly methane. Longer hydrocarbon chains require more oxygen and are therefore more likely to undergo incomplete combustion than shorter hydrocarbon chains. Therefore, natural gas is the least likely fossil fuel to undergo incomplete combustion.

Creativity, activity, service

Strand: Service

Learning outcome: Demonstrate engagement with issues of global significance

The quality of the air that we breathe can have a significant effect on our health. Particulate matter [🔗](https://www.epa.gov/pm-pollution/particulate-matter-pm-basics) (<https://www.epa.gov/pm-pollution/particulate-matter-pm-basics>) (PM) is very small airborne particles such as soot, dust or smoke. Sources of PM include the burning of gasoline or diesel fuel in internal combustion engines and the combustion of wood. PM can be classified as PM₁₀ or PM_{2.5} depending on the size of the particles. Short-term health effects of PM_{2.5} include irritation to the eyes, nose or lungs. Long-term effects include premature death in those people who suffer from pre-existing health conditions.

A possible CAS experience could be to determine the PM_{2.5} levels in your [region](https://www.stateofglobalair.org/air/pm) [🔗](https://www.stateofglobalair.org/air/pm) (<https://www.stateofglobalair.org/air/pm>) and raise awareness of the possible health effects. You could also research ways in which the levels of PM_{2.5} can be reduced and share these with your school or local community.

In the activity below you will use an interactive to help you to make connections between global atmospheric temperatures and atmospheric concentrations of carbon dioxide.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Thinking skills
 - Experimenting with new strategies for learning
 - Combining different ideas in order to create new understandings
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual or pairs activity

Instructions

In this section you have studied the relationship between carbon dioxide emissions and the greenhouse effect. Using the following interactive simulation, you will look at modelling the relationship between atmospheric carbon dioxide concentrations and other dependent variables like atmospheric temperature, oceanic carbon dioxide concentration, and atmospheric water vapour concentrations.

<https://has.concord.org/global-climate.html>  (<https://has.concord.org/global-climate.html>)

1. Create a data table containing space for three different concentrations of atmospheric carbon dioxide (50%, 100% and 150% of 2010 levels).
2. Adjust the level of Human Emissions (% of 2010 values) for carbon dioxide by clicking and moving the button to the left or right. This will allow you to set the three different concentrations of carbon dioxide you chose.
3. Run the simulation for at least a 25-year period.
4. Record the approximate temperature change for the atmosphere, for each concentration of carbon dioxide that you chose.
5. Describe the trend(s) you observe between atmospheric carbon dioxide levels and atmospheric temperature. Support your answer with information from this unit.
6. To what extent does the combustion of fossil fuels relate to the increase in atmospheric temperatures?
7. Examine the relationship that exists between the type of fossil fuel combusted (coal, crude oil, natural gas) and the quantity of carbon dioxide released into the atmosphere.
8. Extension question: Interpret the relationship that is shown in the simulation between carbon dioxide concentrations and oceanic carbon dioxide concentration.